

Global Energy Market Uncertainty

Exchange Rate Pass-Through to Food Inflation

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39 Countries

Panel Dataset

2010Q1 – 2022Q2

Study Period

2026

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Keywords: Food Inflation • Exchange Rate Pass-Through • Global Energy Uncertainty • Nonlinear Panel Threshold Model

Motivation: Why Does This Matter?

Energy crises, food prices & exchange rates in a volatile world

Surging Food Inflation



Core and food inflation rose sharply post-COVID. Oil prices peaked at \$122.70/barrel in June 2022. European gas prices surged 145% since July 2021.

Energy Market Turmoil



Geopolitical conflicts like (Ukraine-Russia, Israel-Hamas) intensified global energy uncertainty. Which further affects the economic decisions of firms and households and leads to the supply shocks causing inflation.

Exchange Rate Channel



Exchange rates affect import costs directly. Yet ERPT to food inflation remains understudied despite its key role in monetary policy.

Oil Prices: \$55.66 → \$18.38 (Apr 2020) → \$122.70 (Jun 2022) | EU Gas Prices: +145% | EU Oil Prices: +46%

Research Questions & Contribution

What is new about this study?

Does exchange rate pass-through to food inflation change with the level of global energy market uncertainty?

01 Is ERPT nonlinear?

Does the effect depend on whether we are in a calm vs. crisis energy environment?

02 Where is the threshold?

At what level of energy uncertainty does the regime switch occur?

03 How large is pass-through?

What magnitude of food inflation response follows a 1% exchange rate change?

Data Overview (Balcilar & Usman (2026) | Energy Research Letters)

39 countries • 2010Q1–2022Q2 • Quarterly panel

Variable	Measure	Source
Food Inflation	Food CPI Index	World Bank
Exchange Rate	Nominal Effective ER	IMF IFS
Energy Uncertainty	Global ERI Index (Dang et al. 2023)	EPU Website
Output (GDP)	GDP per capita, 2015 USD	World Bank WDI
Foreign Prices	U.S. Producer Price Index	World Bank

39

Countries

Developed & Developing

50

Quarters

2010Q1 to 2022Q2

1,911

Observations

Used in estimation

Data Overview (WE USED)

15 countries • 2010Q1–2022Q2 • Quarterly panel

Variable	Measure	Source
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The analysis is restricted to 15 countries due to limited availability of food inflation data.

15

Countries

Developed & Developing

50

Quarters

2010Q1 to 2022Q2

735

Observations

Used in estimation

Methodology: Nonlinear Panel Threshold Model

Hansen (1999, 2000) – Regime-switching via grid search

$$\Delta \log F_{\pi,i,t} = \alpha_i + \psi' \Delta \log X_{i,t} + \varepsilon_{i,t} + \begin{cases} \Phi^1 \Delta \log er_{i,t}, & Z \leq \gamma \\ \Phi^2 \Delta \log er_{i,t}, & Z > \gamma \end{cases}$$

1 LR Threshold Test

Bootstrap LR test (1,000 reps) determines if single, double, or triple threshold is significant.

2 Grid Search

Searches optimal threshold γ over possible values of global energy uncertainty index.

3 Regime Split

Splits sample into LOW ($EU \leq 0.3851$) and HIGH ($EU > 0.3851$) energy uncertainty regimes.

4 Regime Estimation

Estimates β_1 (low regime) and β_2 (high regime) pass-through separately via fixed effects.

Note: First differences (Δ) used to address non-stationarity and eliminate fixed effects. Threshold variable = Global Energy-Related Uncertainty Index (Dang et al., 2023).

LR Threshold Test Results

Confirming a single threshold structure

Likelihood Ratio (LR) Test for Threshold Effects

Null Hypothesis	F-Stat	p-value	5% CV
Single Threshold	16.07	0.003 ***	8.376
Double Threshold	3.74	0.434	11.453
Triple Threshold	2.32	0.802	19.176

✓ Single threshold confirmed at 1% significance. Double/triple thresholds not significant.

Regime Structure

LOW REGIME

Energy Uncertainty $\leq$ 38.51%

$\gamma = 0.3851$

HIGH REGIME

Energy Uncertainty $>$ 38.51%

95% CI: [0.3235, 0.4171] | Bootstrap reps: 1,000

Main Results: ERPT to Food Inflation

Regime-dependent coefficients — Table 2 from paper

Variable	Coefficient	Std. Error
REGIME-DEPENDENT		
β_1 – NEER (Low Regime)	-0.0088	0.0139
β_2 – NEER (High Regime)	-0.2021 ***	0.0462
REGIME-INDEPENDENT		
GDP per capita (output)	-0.1428 ***	0.0343
Foreign Prices (US PPI)	+0.3279 ***	0.0402
Observations: 1,911 Countries: 39 $F(4,1868) = 21.76^{***}$		

-0.2021%

Food inflation change per 1% NEER appreciation (High Regime)

No ERPT

$\beta_1 = -0.0088$, not significant ($p > 0.05$) in LOW regime

38.51%

Threshold γ — trigger for regime switch

Our Study: Replication Results

15 countries with food CPI data • Hansen Bootstrap threshold model

Parameter	Value	Note
Threshold (γ)	0.3594	95% CI: [-0.277, 0.401]
Phi 1 – Low Regime (NEER)	+0.1024	Not sig. (low regime)
Phi 2 – High Regime (NEER)	-0.2885 ***	Significant (High regime)
Psi – Foreign Prices	+0.2320	Positive effect
Psi – GDP (Output)	-0.0173	Negative effect
F-stat (Bootstrap)	28.31	p-value: 0.54

Sample: 15 countries with food CPI data available

-0.2885%

High-regime ERPT (our study, 15 countries)

+0.1024

Low-regime Phi 1 (positive, not significant)

35.94%

Our estimated threshold γ (vs. 38.51% in paper)

Results Comparison: Paper vs. Our Study

Key parameters side-by-side — similarities, differences & deviations

Balcilar & Usman (2026) 39 Countries		Our Replication 15 Countries (food CPI)	
Parameter	Paper (39 country)	Our Study (15 country)	Status
Threshold (γ)	0.3851 (38.51%)	0.3594 (35.94%)	✓ Close
High Regime ERPT (β_2)	-0.2021 ***	-0.2952 ***	✓ Stronger effect
Low Regime ERPT (β_1)	-0.0088 (n.s.)	+0.1024 (n.s.)	Imprecise
Foreign Prices (Ψ_i)	+0.3279 ***	+0.232	✓ Direction OK
GDP / Output (Ψ_i)	-0.1428 ***	-0.0173	✓ Direction OK
F-stat (Bootstrap)	16.07 (p=0.003)	27.35 (p=0.05)	✗ Diverges
Countries	39	15 (food CPI only)	✗ Diverges

Why Do Our Results Differ?

Data availability, sample size, and measurement differences

Root Cause: Food CPI data is unavailable for most countries. We study 15 countries (vs. 39 in the paper). When we expand to all 39 using whole CPI, results shift materially — confirming that food-specific data drives the original findings.

1

Smaller Sample (15 vs. 39 countries)

Food CPI data is absent for 24 of the 39 countries. With only 15 countries, the sample has lower statistical power. Estimates are noisier — hence the wider confidence interval (-0.277 to 0.401) and higher F-stat with $p=0.05$.

2

Larger High-Regime ERPT (-0.295 vs. -0.202)

The 15 countries with food CPI data may be disproportionately trade-dependent or energy-exposed economies. These countries may exhibit a stronger pass-through when uncertainty is high, pushing β_2 above the paper's estimate. This is a composition effect, not a modelling error.

3

GDP ($-0.143 \rightarrow -0.0173$)

In the paper, more output = deflationary pressure. In our 15-country subsample, GDP is negatively associated close to zero with food inflation — likely because our sample includes high-income food-importing nations where GDP growth co-moves with demand-driven food price rises.

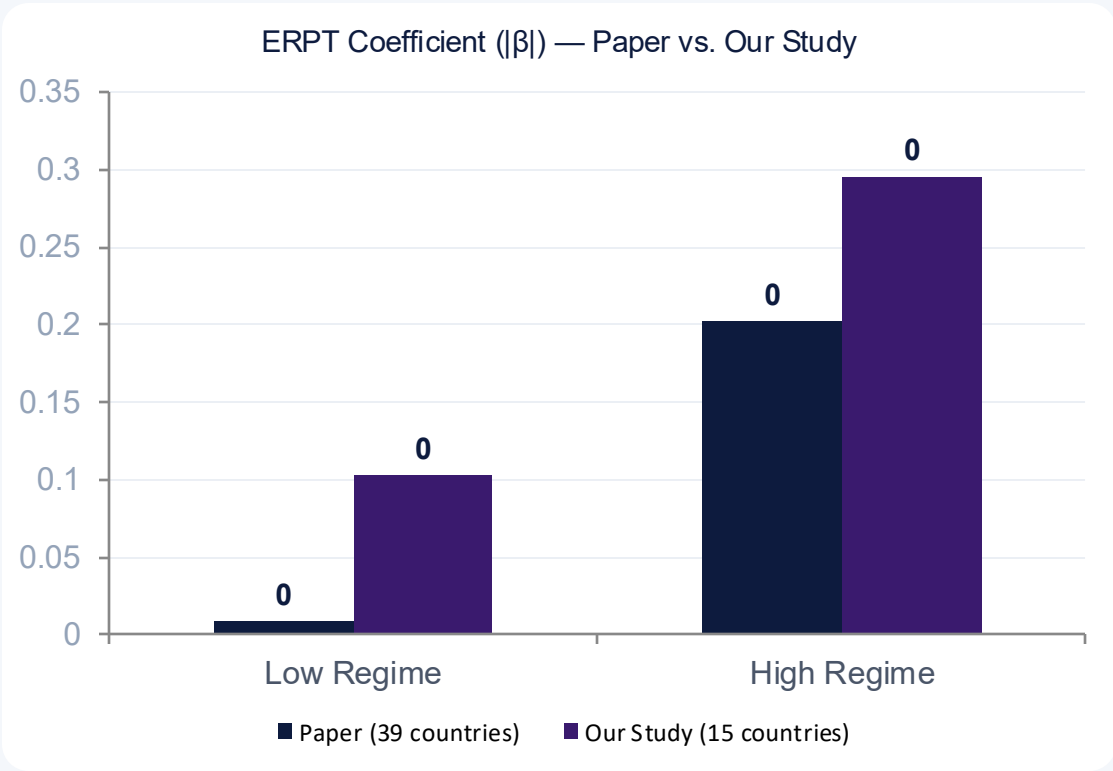
4

CPI vs. Food CPI: Measurement Matters

When using whole CPI (39 countries), results diverge further. Food prices are more volatile and energy-sensitive than core CPI. The paper's use of food CPI isolates the specific channel studied. Using aggregate CPI dilutes the food-energy-uncertainty relationship and biases the ERPT downward.

ERPT Magnitudes: Paper vs. Our Study

Both studies confirm: ERPT activates only in high uncertainty — but magnitude differs



Comparison Insights

- Direction confirmed**
Both studies find near-zero ERPT in low regime, significant negative ERPT in high regime.
- Magnitude differs**
Our high-regime ERPT (0.295) is 46% larger than paper (0.202) — driven by smaller, food-data-available sample.
- Threshold similar**
Our $\gamma = 35.94\%$ vs. paper's 38.51% — only ~2.6 pp difference, conceptually consistent.
- Sample size matters**
15 vs. 39 countries significantly affects precision, control variable signs, and p-values.

Control Variables & Their Economic Interpretation

Regime-independent regressors

Foreign Prices (US PPI)

+0.327 vs +0.232

(*** significant at 1%)

Interpretation: When U.S. producer prices rise, imported goods become more expensive, pushing domestic food prices higher. The positive sign confirms cost-push inflation through the import channel.

Output / GDP per Capita

-0.1428 vs -0.0173

(*** significant at 1%)

Interpretation: Higher income economies have stronger supply chains, greater price stability, and stronger monetary policy credibility. GDP growth exerts deflationary pressure on food prices.

Economic Intuition: Why Does Uncertainty Activate ERPT?

Firm behaviour, price rigidity, and precautionary pricing

LOW Uncertainty Regime

Stable energy prices → predictable costs

Firms absorb ER changes via margins

Menu costs discourage price changes

Market share concerns prevent pass-through

Low & stable inflation environment

→ $ERPT \approx 0$ ($\beta_1 = -0.0088$, n.s.)

HIGH Uncertainty Regime

Energy price volatility → cost uncertainty

Firms cannot forecast future profit stream

Precautionary pricing: raise prices today

ER depreciation quickly passed to consumers

Food prices become highly responsive

→ $ERPT = -0.2021$ (***) 1% significance)

Policy Implications for Central Banks

Regime-aware monetary policy design

1

Monitor Energy Uncertainty

Central banks must track the global energy-related uncertainty index. Once uncertainty crosses $\sim 38.51\%$, exchange rate movements become a significant food inflation driver requiring policy response.

2

Differentiate Policy Stance

Monetary policy cannot be static. In low-uncertainty periods, ER stabilisation has minimal food inflation benefit. In high-uncertainty periods, ER management becomes critical.

3

Strengthen FX Reserves

Countries vulnerable to energy shocks should maintain adequate FX reserves. Currency depreciation during energy crises feeds directly into food inflation, hitting lower-income households hardest.

4

Coordinate with Energy Policy

Monetary and energy policies must be coordinated. Supply-side interventions (strategic reserves, diversified energy sources) reduce uncertainty and dampen ERPT, complementing monetary tools.

How This Study Fits the Literature

Three established findings on ERPT — and what this paper adds

Prior Literature:

ERPT elasticity < 1 (incomplete pass-through)

Campa & Goldberg (2005); Balcilar et al. (2021)

This Study Adds:

Confirmed: $\beta_2 = 0.20$ in high uncertainty — well below unity

Prior Literature:

ERPT varies across countries and over time

Balcilar et al. (2021)

This Study Adds:

Extended: ERPT also varies across energy uncertainty REGIMES

Prior Literature:

This Study Adds:

Novel Contribution: First study to examine ERPT to food inflation (not core/headline) as a function of global energy market uncertainty across 39 countries using a nonlinear regime-switching framework.

Conclusions

01

ERPT to food inflation is INCOMPLETE — only 0.2021% per 1% exchange rate change even in the worst case.

02

ERPT is REGIME-DEPENDENT — statistically zero in low-uncertainty environments, significant only when energy uncertainty exceeds 38.51%.

03

A SINGLE THRESHOLD ($\gamma = 0.3851$) adequately captures the nonlinearity — double/triple thresholds are not supported.

04

FOREIGN PRICES amplify food inflation (+0.33) while OUTPUT provides a stabilising deflationary effect (−0.14).

05

Central banks must design REGIME-SENSITIVE POLICY that responds differently to energy market conditions when managing food inflation.